Threshold Phases and the Limits of Motion Compensation for Event Streams

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***Abstract*—An event camera reports when the log intensity of a pixel crosses a threshold. When an edge moves across the sensor, every pixel on its path sees the same intensity profile, yet the pixels do not emit the same events: each fires on its own lattice of threshold levels, offset by a threshold phase, and every event carries timing jitter. We determine the rate that these differences require in an analytic model: a ramp edge that passes each pixel once under an affine flow, with independent uniform phases and Gaussian jitter. Given the edge and its motion, a lossless code must still describe the phase and the jitter of every pixel, so its rate per event does not fall as the path grows. Under a mean-squared timing tolerance the rate has three regimes. Below the jitter variance, the code describes every event time, and the rate is exact. Above it, the code describes one number per pixel, and the rate is known within 0.255 bit. Beyond a threshold set by the event spacing of the pixel, the rate given the edge and its motion is zero, and a description of the motion needs a number of nats that grows at most logarithmically with the length of the path. A sender has the events and not the edge, and a common substitute is to move reference events along the motion. A moved event keeps the phase of its source pixel, so on a voxel grid the counts moved along the true motion reproduce only a share of about Δ_t/τ of the true counts when the time bin Δ_t is shorter than the event spacing τ. On recordings from a static camera and from a moving car, the moved events fall at uniform positions within the event trains of their target pixels, as independent phases imply, for every pixel spacing below 4 ms; they lock only in the band of mains lighting. For a standard HEVC coder on count pictures, this means that on fine time grids its P pictures do not undercut a list of the counts, and its lossy errors are single events that it removes, so that intra pictures reach a given fidelity with 0.69 to 1.00 times the bits of P pictures. The ratio of the time bin to the event spacing tells a sender whether motion-compensated prediction of counts can pay.**

***Index Terms*—Event camera, send-on-delta sampling, level-crossing sampling, rate-distortion theory, generalized sampling, motion compensation, video coding, HEVC, neuromorphic sensing.**

I. Introduction

An event camera has no frames. Each pixel compares its log intensity with a stored reference and emits an event, with a time stamp and a polarity, whenever the difference reaches a contrast threshold [1], [2]. The output is a stream of address-events whose rate follows the scene: it is close to zero for a static scene and rises with the speed and the contrast of what moves. A sensor node that forwards this stream over a rate-limited link must decide how many bits the stream needs.

The stream of a moving scene looks highly redundant. If an object moves across the sensor, the pixels along the path of one of its edges observe the same intensity profile, one after the other. Yet they do not emit the same events. Each pixel compares the intensity with its own reference, so it fires on its own lattice of levels, and the offset of that lattice, the threshold phase of the pixel, differs from pixel to pixel. The same profile therefore produces a different event train at every pixel (Fig. 1). This paper asks what these differences cost: how many bits a code needs to preserve them, and how many it saves when a timing error is allowed.

The setting has precedents in sampling theory. An event pixel is a send-on-delta sampler [3], which for a monotone input coincides with level-crossing sampling [4], [5]. An object that crosses M pixels is sampled by M such samplers, which recalls the generalized sampling expansion [6], by which a bandlimited signal is recovered from the outputs of m linear systems that are each sampled at $1/m$ of the Nyquist rate. In multi-channel time encoding, several integrate-and-fire encoders with shifted integrators share one input, and the shifts are what make the channels complementary [7]–[9]. The threshold phases play that role here.

Existing event coders approach the rate empirically. Lossless coders exploit the spatial and temporal correlation of addresses and time stamps [10]–[12]. Lossy coders discard or aggregate events [13], [14], and the coder of [15] suppresses events that an on-sensor optical flow can predict and lets the receiver extrapolate them. A recent survey notes that reported results are hard to compare [16], and no accepted distortion measure for lossy event coding exists [17]. On the theoretical side, rate-distortion results for point processes concern Poisson and finite point processes [18]–[21]. They do not model events that a moving profile generates through a threshold rule. Event streams have also been aggregated into count pictures per polarity and coded without loss [11] by HEVC [22], a video coder whose prediction is motion compensated [23], but what the prediction saves on such pictures has not been reported.

This paper models that mechanism and derives what a code can and cannot save. The phase mechanism holds for any scene (Lemma 1). For the rates we specialize to a ramp edge that passes each pixel once under an affine flow, with independent uniform phases and Gaussian jitter, which isolates the mechanism and gives closed forms. Recordings then test its premise, and a standard video coder shows its consequences. The central result is the rate-distortion function of the events of a pixel under a timing tolerance. The other results lead to it or follow from it.

1) Mechanism. Given the edge and its motion, the events of a pixel are a function of a single number, its threshold phase, up to timing jitter. The phase is a property of the pixel and is independent across pixels. Next to what the pixels of a path share, the edge and its motion, each pixel therefore holds a private quantity. A lossless code must describe it for every pixel, and its rate per event does not decrease with the number M of pixels on the path (Theorem 1).

2) Rate under a timing tolerance. The tolerance decides which of the private quantities a code must describe (Theorem 2). Below the jitter variance it describes all of them, and the rate is known exactly. Above it one number per pixel remains, and the rate is known within 0.255 bit. Beyond a threshold the rate given the edge and its motion is zero: the decoder regenerates the events. The threshold grows with the event spacing of the pixel, which is the contrast threshold divided by the rate of change of its intensity, so slowly crossed pixels need a larger tolerance. Along a path of M pixels, a description of the motion of $O(\ln M)$ nats then replaces a lossless description of order M.

3) Moved events, on recordings and in a video coder. A sender has the events and not the edge. A common substitute, and the one that the prediction of a video coder implements, is to move reference events along the motion. A moved event keeps the phase of its source pixel. At equal event spacings this doubles the timing error of the ideal decoder (Proposition 2), and on a voxel grid the counts moved along the true motion reproduce a share of about Δ_t/τ of the true counts when the bin is shorter than the event spacing. On recordings from a static camera [24] and from a moving car [25], a phase test shows that the premise holds: moved events fall at uniform positions within the trains of their target pixels. We then code count pictures of the same recordings with x265, an unmodified HEVC encoder. On fine time grids its P pictures do not undercut a list of the counts, its lossy errors are removed single events, and intra pictures reach a given fidelity with as few bits as P pictures or fewer.

Section II states the model. Sections III and IV contain the theory, Section V the moved events, and Section VI the measurements. Proofs are in the appendices. Rates are in nats unless bits are named, $\|\cdot\|$ is the Euclidean norm of a vector and the spectral norm of a matrix, and $[x]^+ = \max(x, 0)$.

II. Model

We first state the pixel rule, from which the phase mechanism follows for any scene. We then specialize to a ramp edge in uniform passage, for which the rates have closed forms.

A. Pixel

Pixels sit at the lattice points $i \in \mathbb{Z}^2$ of the image plane. Pixel i observes the log intensity $\ell_i(t) = L(i,t)$ of a continuous field L.

(A1) Send-on-delta. Pixel i holds a reference μ_i with $|\ell_i - \mu_i| < C$ at the start. It emits an event at the first time t at which $|\ell_i(t) - \mu_i| = C$, with polarity $p = \operatorname{sign}(\ell_i(t) - \mu_i)$, and then sets $\mu_i \leftarrow \mu_i + pC$. The contrast threshold $C > 0$ is common to all pixels and known to the decoder. The *phase* of the pixel is the number $a_i \in (0, C]$ with $\mu_i \in a_i + C\mathbb{Z}$ at the start.

B. Scene and Motion

A template $g : \mathbb{R}^2 \to \mathbb{R}$ gives the log intensity in object coordinates. At time t the object point u is at the image position $\phi_t(u) = J_t u + b_t$, with $J_t \in \mathbb{R}^{2\times 2}$ invertible, so

$$L(x,t) = g\big(\phi_t^{-1}(x)\big) \tag{1}$$

and pixel i observes

$$\ell_i(t) = g\big(u_i(t)\big), \qquad u_i(t) = J_t^{-1}(i - b_t). \tag{2}$$

The curve u_i is the trajectory of pixel i in object coordinates. The velocity field $v(x,t) = \dot\phi_t(\phi_t^{-1}(x))$ is affine in x and satisfies the brightness constancy relation

$$\frac{\partial L}{\partial t}(x,t) + \langle \nabla L(x,t), v(x,t)\rangle = 0, \tag{3}$$

which follows from differentiating $L(\phi_t(u), t) = g(u)$ in t. We call the motion an *affine flow* when the velocity field is also constant in time,

$$v(x) = \Lambda x + v_0, \tag{4}$$

with generator $\Lambda \in \mathbb{R}^{2\times 2}$. A translation has $\Lambda = 0$. The trace of Λ is the rate at which the object grows or shrinks in area, its antisymmetric part rotates the object, and its symmetric traceless part shears it. At a given instant, any smooth velocity field has the form (4) to first order in x around a point, so (4) is the local model of a general motion over a short time.

C. Edge

(A2) Ramp edge. The template is a ramp across a line with unit normal ν,

$$g(u) = \varrho\big(\langle \nu, u\rangle - c_0\big), \qquad \varrho(z) = \begin{cases} 0, & z \le 0, \\ \gamma z, & 0 \le z \le w, \\ V, & z \ge w, \end{cases} \tag{5}$$

with slope $\gamma > 0$, width w and height $V = \gamma w = nC$ for an integer $n \ge 1$.

Pixel i sees the edge through its edge coordinate

$$z_i(t) = \langle \nu, u_i(t)\rangle - c_0, \qquad \ell_i(t) = \varrho\big(z_i(t)\big). \tag{6}$$

(A2′) Uniform single passage. The edge passes pixel i once: $z_i < 0$ before a time t_i, $z_i > w$ after the passage, and in between z_i is affine in t with slope $r_i > 0$.

Inside the ramp, $\nabla L = \gamma \nu_t$ with $\nu_t = J_t^{-\top}\nu$ a normal of the edge in the image, and $\dot z_i = -\langle \nu_t, v(i,t)\rangle$. By (3), the intensity of pixel i rises during the passage at the rate

$$\gamma r_i = -\langle \nabla L, v\rangle = \|\nabla L\|\, v_{\perp,i}, \tag{7}$$

with the field evaluated at pixel i. Here $v_{\perp,i} = -\langle \nabla L, v\rangle/\|\nabla L\|$ is the speed at which the edge advances over the pixel along its normal, counted positive when the intensity of the pixel rises, as it does in a passage with $r_i > 0$. We call

$$\tau_i = \frac{C}{\gamma r_i} = \frac{C}{\|\nabla L\|\, v_{\perp,i}} \tag{8}$$

the *event spacing* of pixel i. It is the time that the ramp takes to pass the pixel, divided by n. At a fixed image slope $\|\nabla L\|$ it is inversely proportional to the normal speed of the edge. Assumption (A2′) holds exactly for a translation, where τ_i is the same for all pixels, and for a plane that approaches the camera along the optical axis at constant speed, where J_t^{-1} is affine in t and b_t is fixed. For an affine flow it holds to first order (Remark 1), and τ_i varies from pixel to pixel.

D. Phase, Jitter and Clock

(A3) Phase. Before the edge arrives, pixel i rests at the level 0 with its reference behind the passage, $\mu_i \in (-C, 0]$, so that $a_i = \mu_i + C$. The phases a_i are independent across pixels, uniform on $(0, C]$, and independent of the jitter.

(A4) Jitter and clock. The kth event of pixel i is recorded at $T_{i,k} = t_{i,k} + \varepsilon_{i,k}$, where $t_{i,k}$ is the time given by (A1) and the $\varepsilon_{i,k}$ are independent $\mathcal{N}(0, \sigma^2)$ variables, independent of the phases, with $\sigma > 0$. The stored value is $Y_{i,k} = \delta\lfloor T_{i,k}/\delta \rfloor$ for a clock resolution δ. The index k of each recorded event is known.

The uniform phase does not arise inside the model, in which every pixel rests at the same level. It stands for the offsets of real pixels and for their earlier activity. A pixel that the edge crosses from the high level to the low level is covered by the mirrored assumption, with its reference in $[V, V + C)$. Remark 4 treats a reference that is ahead of the passage.

We write $T_i = (T_{i,1}, \dots, T_{i,n})$ and $Y_i = (Y_{i,1}, \dots, Y_{i,n})$, and $\mathcal{P}$ for the set of pixels that the edge passes during the observation, with $M = |\mathcal{P}|$. Let $\eta(a, \sigma)$ denote the differential entropy of the sum of a uniform variable on $(0, a)$ and an independent $\mathcal{N}(0, \sigma^2)$ variable. It satisfies

$$\tfrac{1}{2}\ln\big(a^2 + 2\pi e\sigma^2\big) \le \eta(a, \sigma) \le \tfrac{1}{2}\ln\big(2\pi e(a^2/12 + \sigma^2)\big) \quad (9)$$

by the entropy power inequality [26, Ch. 17] and by the Gaussian bound on the entropy at a given variance [26, Ch. 8], and $\eta(a, \sigma) - \ln a \to 0$ as $\sigma/a \to 0$.

III. Structure and Lossless Limit

Lemma 1. *Under (A1), for any continuous ℓ_i, every event of pixel i occurs at a time t with $\ell_i(t) \in a_i + C\mathbb{Z}$.*

The lattice $a_i + C\mathbb{Z}$ never changes. The phase is a property of the pixel, set by its history and not by the edge that passes.

Lemma 2. *Under (A1), (A2), (A2′) and $\mu_i \in (-C, 0]$ before the passage, pixel i emits exactly n events, all of positive polarity, at*

$$t_{i,k} = t_i + A_i + (k-1)\tau_i, \qquad k = 1, \dots, n, \quad (10)$$

where $A_i = a_i/(\gamma r_i)$. Under (A3), A_i is uniform on $(0, \tau_i]$.

Given the template and the flow, the noise-free events of a pixel are thus a function of one number, its phase. Fig. 1 shows two pixels with $n = 3$ that see the same ramp: the intensity trajectories coincide, and the event trains differ by the difference of the phases. In the object coordinate, an event is a point of the graph of the profile on the level lattice of its pixel: the M pixels of a path sample the same profile on M lattices with independent offsets.

Theorem 1 (Lossless code length)**.** *Under (A1) to (A4) and (A2′), the vectors T_i, $i \in \mathcal{P}$, are independent given the template and the flow,*

$$h(T_i) = \frac{n-1}{2}\ln(2\pi e\sigma^2) + \eta(\sqrt{n}\,\tau_i, \sigma), \quad (11)$$

and

$$H(Y_i \mid g, \phi) = n\ln\frac{1}{\delta} + h(T_i) + e_\delta, \quad (12)$$

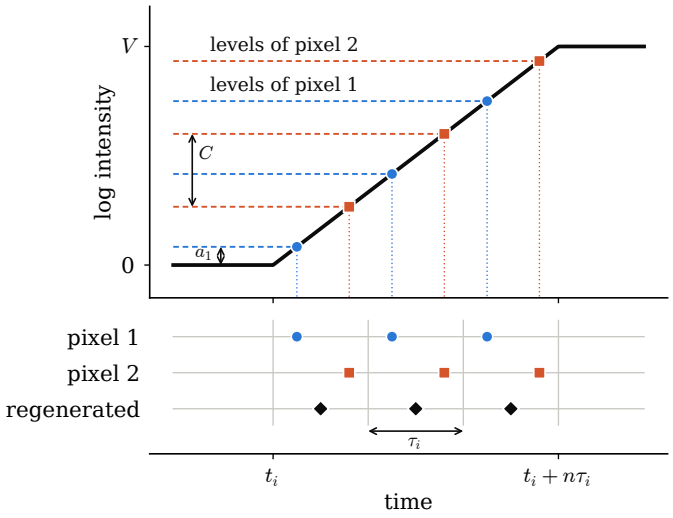


Fig. 1. One ramp, two pixels with different threshold phases, and the regenerated events (22). Time is measured from the arrival of the edge at each pixel. Jitter is omitted.

where $e_\delta \ge 0$ for every δ, and $e_\delta \to 0$ as $\delta/\sigma \to 0$, uniformly in τ_i.

When $\sigma \ll \tau_i$ and $\delta \ll \sigma$, (12) gives the cost per event

$$\frac{H(Y_i \mid g, \phi)}{n} \approx \frac{n-1}{n}\ln\frac{\sigma\sqrt{2\pi e}}{\delta} + \frac{1}{n}\ln\frac{\sqrt{n}\,\tau_i}{\delta}. \quad (13)$$

One event of each pixel carries the cost of the phase, and the other $n-1$ carry that of the jitter. The code length of the path is the sum of (12) over $\mathcal{P}$. It grows in proportion to M, and neither term of (13) falls as the path gets longer: a decoder that knows the edge and its motion exactly still needs (13) for every event. What the pixels of a path share, the edge and its motion, a code can describe once. What is private to each pixel, its phase and its jitter, it must describe for each pixel, and (12) is that irreducible conditional rate. Section VI-E measures what the motion saves a video coder on recordings.

Remark 1 (Accuracy of the uniform passage). For an affine flow (4), $\dot{\nu}_t = -\Lambda^\top \nu_t$ and $\ddot{z}_i = \langle \nu_t, \Lambda v(i)\rangle$, hence $|\ddot{z}_i| \le \kappa_i |\dot{z}_i|$ with $\kappa_i = \|\Lambda\|/|\cos\vartheta_i|$, where ϑ_i is the angle between the velocity $v(i)$ and the normal ν_t. Take r_i as the slope of z_i at t_i. Over a passage of duration $P_i = n\tau_i$, the event times then deviate from (10) by at most $\kappa_i P_i^2/2$, to first order in $\kappa_i P_i$. For a generator of norm $0.3\,\mathrm{s}^{-1}$, a passage of 5 ms and motion along the normal this is $4\,\mu$s, three orders of magnitude below the passage time. The bound grows without limit for edges that are nearly parallel to their motion. Such edges pass few pixels per unit time and contribute few events.

Remark 2 (Several edges). Theorem 1 counts one edge that passes each pixel once. If further edges of the same object pass the same pixel, they meet the same lattice, so the phase is paid once per pixel and not once per edge. With many edges per pixel the cost per event tends to the jitter term $\ln(\sigma\sqrt{2\pi e}/\delta)$, which does not depend on the path either.

IV. Rate Under a Timing Tolerance

The results of this section and the next assume decoders with different inputs. Theorems 2 and 3 count the bits of the event times when the encoder and the decoder know the template, the flow, and hence the pixels that the edge passes and the number of events of each, and when the index of every event is known (A4). Their zero rate is a rate given all of this. Proposition 1 adds the cost of the flow, while the decoder still holds the template. A practical sender holds neither, and Section V turns to what it can send instead.

The decoder outputs $\hat{T}_i \in \mathbb{R}^n$ for each pixel, and the distortion is the mean-squared timing error

$$d(T_i, \hat{T}_i) = \frac{1}{n}\|T_i - \hat{T}_i\|^2. \tag{14}$$

A. One Pixel

Let $R_i(D) = \inf I(T_i; \hat{T}_i)$, with the infimum over the conditional laws of $\hat{T}_i$ that satisfy $\mathbb{E}\, d(T_i, \hat{T}_i) \le D$, be the rate-distortion function of the events of pixel i. It is the rate that block codes over many independent passages approach. Define

$$s_i^2 = \frac{n\tau_i^2}{12} + \sigma^2, \qquad D_{A,i} = \sigma^2 + \frac{\tau_i^2}{12}, \tag{15}$$

$$\Gamma_i = \tfrac{1}{2}\ln(2\pi e\, s_i^2) - \eta(\sqrt{n}\,\tau_i, \sigma). \tag{16}$$

By (9), $0 \le \Gamma_i \le \frac{1}{2}\ln(2\pi e/12)$, which is 0.255 bit, for every σ and τ_i.

Theorem 2 (Rate-distortion function of a pixel)**.** *Under (A1) to (A4) and (A2′),*

$$R_i(D) = h(T_i) - \frac{n}{2}\ln(2\pi e D) \qquad \textit{for } 0 < D \le \sigma^2, \tag{17}$$

$$\big[U_i(D) - \Gamma_i\big]^+ \le R_i(D) \le U_i(D) \qquad \textit{for } \sigma^2 < D < D_{A,i}, \tag{18}$$

$$R_i(D) = 0 \qquad \textit{for } D \ge D_{A,i}, \tag{19}$$

with

$$U_i(D) = \frac{1}{2}\ln\frac{s_i^2}{nD - (n-1)\sigma^2}. \tag{20}$$

Moreover,

$$R_i(D) \ge \frac{n\,(D_{A,i} - D)}{2s_i^2} \qquad \textit{for } 0 < D \le D_{A,i}, \tag{21}$$

so that $R_i(D) > 0$ for $D < D_{A,i}$.

The three regimes have a direct reading. Below the jitter variance, every bit of timing precision costs a bit, per event: the rate falls by $\log_2 10 = 3.32$ bits per event and per decade of the root-mean-square tolerance $\sqrt{D}$. Between σ^2 and $D_{A,i}$, the $n-1$ jitter components orthogonal to a common shift of the events are no longer described, and the code describes one number per pixel, the phase together with the mean jitter, to within 0.255 bit. From $D_{A,i}$ on, nothing is described. The decoder that attains (19) outputs the mean times

$$\hat{T}_{i,k} = t_i + \big(k - \tfrac{1}{2}\big)\tau_i, \qquad k = 1, \dots, n. \tag{22}$$

It *regenerates* the events of the pixel from the edge and the flow. Each regenerated event is within $\tau_i/2$ of its noise-free time, and the count and the polarities are exact.

Two of these statements extend beyond the ramp. If the noise-free event times of a pixel are any function of its phase, (17) holds below the jitter variance with the entropy of the actual event times, and the smallest distortion at zero rate is $\frac{1}{n}\operatorname{tr}\operatorname{Cov}(T_i)$, attained by the mean times. The middle regime, in which one number per pixel is described to within 0.255 bit, and the closed forms (11), (20) and (15) belong to the ramp in uniform passage, where the phase shifts all events of a pixel by a common time.

The threshold $D_{A,i}$ grows with the square of the event spacing. By (8), a pixel whose intensity changes slowly has a large spacing and needs a large tolerance before its events can be regenerated, and a pixel that is crossed quickly needs a small one.

B. A Path

For the M pixels of a path, let the distortion be the average of (14) over $i \in \mathcal{P}$, and let $R_{\mathcal{P}}(D)$ be the rate-distortion function of $(T_i)_{i\in\mathcal{P}}$ under it.

Theorem 3 (Rate-distortion function of a path)**.** *Under (A1) to (A4) and (A2′),*

$$R_{\mathcal{P}}(D) = \min\Big\{ \sum_{i\in\mathcal{P}} R_i(D_i) : \ \frac{1}{M}\sum_{i\in\mathcal{P}} D_i \le D \Big\}. \tag{23}$$

In particular,

$$R_{\mathcal{P}}(D) = \sum_{i\in\mathcal{P}} h(T_i) - \frac{Mn}{2}\ln(2\pi e D) \qquad \textit{for } 0 < D \le \sigma^2, \tag{24}$$

and $R_{\mathcal{P}}(D) = 0$ if and only if

$$D \ \ge \ \bar{D}_A = \sigma^2 + \frac{1}{12M}\sum_{i\in\mathcal{P}} \tau_i^2. \tag{25}$$

If $\tau_i \le \tau_{\max}$ on the path, then for $D \le \bar{D}_A$

$$R_{\mathcal{P}}(D) \ge \frac{nM\,(\bar{D}_A - D)}{2\,(n\tau_{\max}^2/12 + \sigma^2)}. \tag{26}$$

If the tolerance is imposed on every pixel separately, the rate is zero if and only if $D \ge \max_i D_{A,i}$, and the slowest crossing sets the tolerance. By (26), below $\bar{D}_A$ a rate that is proportional to M remains.

C. The Motion

Theorems 2 and 3 assume that the decoder knows the flow. It remains to count what describing the flow costs. An error in a speed displaces the regenerated events in proportion to the distance that the edge has traveled, so a longer path needs the motion to a finer precision for the same distortion. A parameter of fixed dimension costs only logarithmically more at a finer precision, and Proposition 1 states this in general form.

Proposition 1 (Description of the motion)**.** *Let the flow depend on a parameter $\theta = (\theta_1, \dots, \theta_q)$ that ranges over a box whose jth side has length at most $L_j M^{\alpha_j}$, and let the mean times*

$\bar{t}_{i,k}(\theta) = t_i(\theta) + (k - \frac{1}{2})\tau_i(\theta)$ *of* (22) *satisfy* $|\partial \bar{t}_{i,k}/\partial \theta_j| \le S_j M^{\beta_j}$ *on the box, for all events of the path, with constants* $L_j, S_j > 0$ *and exponents* $\alpha_j, \beta_j \ge 0$ *that do not depend on* M*. Then for every* $\xi > 0$ *the events of the path can be described with average distortion at most* $(1+\xi)\bar{D}_A$ *using*

$$B \le \Big(\sum_{j=1}^{q} (\alpha_j + \beta_j) \Big) \ln M + O(1) \quad \textit{nats,} \tag{27}$$

where the $O(1)$ *term depends on q,* ξ*,* σ *and the constants and not on* M*.*

For a translation along a line with a speed in a fixed interval that excludes zero, θ holds the position of the edge, with a range of order M and a sensitivity of order one, and the speed, with a range of order one and a sensitivity of order M, so $B \le 2 \ln M + O(1)$. The same holds with finite exponents, and hence $B = O(\ln M)$, for an affine flow whose normal speed is bounded away from zero on the path and whose duration is polynomial in M. The lossless cost of the same events is the sum of (12), of order M, so the ratio of the two grows at least as $M/\ln M$. Above $\bar{D}_A$ the description that remains is $O(\ln M)$. Below it, a rate proportional to M remains. The bound (27) is for a receiver that holds the edge.

Remark 3 (Event count under a change of scale). An edge segment emits n events for every pixel that it passes, so its event rate is n times the area that it sweeps per unit time, that is n times its length times its mean normal speed. Under a similarity flow about a fixed point, growth by a factor ζ makes the segment ζ times longer and ζ times faster, and the rate grows as ζ^2. For an object that approaches at constant speed, the image speed itself grows as ζ^2 and the rate as ζ^3. A regeneration that moves a fixed set of events keeps their number. Under a change of scale it must also change the number of events, or regenerate them from the edge.

Remark 4 (Departures from the model). *Reference ahead of the passage.* If $\mu_i \in (0, C)$ before a rising passage, as after a passage of the opposite polarity at the same pixel, the first level that the intensity meets is the reference itself, which does not fire. The pixel then emits $n-1$ events, and the results hold with $n-1$ in place of n and $t_i + \tau_i$ in place of t_i. *Height.* If the height of the edge is not a multiple of the threshold, $V/C = n + f$ with $0 < f < 1$, pixel i emits $n+1$ events when $a_i \le fC$ and n otherwise. Reproducing the count exactly costs the binary entropy of f per pixel, and a zero-rate statement then holds only for a distortion measure that tolerates one unmatched event per pixel, with a modified threshold. *Sensor.* Noise events, threshold mismatch between pixels, a refractory time and a jitter that depends on the slope are outside the model, and their effect on the three regimes is not analyzed here.

V. Moved Events

The decoder (22) needs the edge and the flow. A sender has neither. It has the events. It could estimate the edge from them, and this paper does not pursue that route. We study a common substitute: the receiver moves reference events that it already holds along the motion. This is motion-compensated prediction, the core of a hybrid video coder [22], [23]: a block of a decoded picture, displaced by a motion vector, predicts a block of the next picture, and the coder sends the residual. This section gives what a moved event costs in timing error and in the fidelity of a receiver that counts events.

A. Timing Error

Proposition 2 (Moved events)**.** *Let the same edge pass pixel* j *in an earlier window and a different pixel* i *later, under (A1) to (A4) and (A2′), and let the motion be exact, so that the* k*th event of pixel* j *is moved to pixel* i *at the time* $\tilde{T}_{i,k} = T_{j,k} + t_i - t_j$*. Then*

$$\frac{1}{n}\, \mathbb{E}\|T_i - \tilde{T}_i\|^2 = D_{A,i} + D_{A,j} + \frac{4n^2 - 1}{12}\, (\tau_i - \tau_j)^2. \tag{28}$$

With equal event spacings the error is $2D_A$, twice that of (22): a moved event carries the phase and the jitter of its source pixel, and both are independent of those of its target. Unequal spacings, as under a change of scale, add the last term, which grows with the square of the number of events per pixel. If the moved events fall on their own pixel, as for an object at rest, the phases coincide and only the jitter term $2\sigma^2$ remains.

B. Voxel Fidelity

The timing error (14) presupposes that predicted and true events are paired. Receivers of event streams count events instead, on a voxel grid: the number of events of each polarity in blocks of $\Delta_x \times \Delta_x$ pixels and in time bins of length Δ_t. A video coder codes the same counts as pictures. For a reconstruction with voxel counts c' and a true set with counts c, of sizes N' and N, the fidelity is

$$F = \frac{2 \sum \min(c', c)}{N' + N} = 1 - \frac{\sum |c' - c|}{N' + N}, \tag{29}$$

the share of the events of the two sets that can be paired one to one inside a voxel.

For blocks of one pixel, a train that is long against the event spacing, and a jitter that is small against the bin, Lemma 2 gives the fidelity of the regenerated events (22) directly: a bin holds at most one event of each train when $\Delta_t \le \tau_i$, the offset between the two trains is uniform, and in expectation over the phase

$$F = \frac{\Delta_t}{\tau_i} \qquad \text{for } \Delta_t \le \tau_i, \tag{30}$$

with $F = 1$ at $\Delta_t = \tau_i$. The voxel counts of a pixel are thus reproduced only when the bin is at least as long as its event spacing, the same scaling as the threshold of Theorem 2, for which $\sqrt{D_{A,i}} = \tau_i/\sqrt{12}$ when $\sigma \ll \tau_i$.

Moved events do almost as well on this grid. With equal spacings and no jitter, the kth moved event lies at the offset $\delta = A_i - A_j$ from the kth true event, and δ has the triangular density $(\tau_i - |\delta|)/\tau_i^2$ on $(-\tau_i, \tau_i)$. Two events at the offset x share a bin with probability $[1 - |x|/\Delta_t]^+$ over the position of the bins. When $|\delta| > \tau_i - \Delta_t$, a moved event can share its bin with a neighbor of its true event instead, which has a partner

for $n-1$ of the n events of a train. For a train of n events and $\Delta_t \le \tau_i/2$, the two contributions give

$$F = \frac{\Delta_t}{\tau_i} - \frac{\Delta_t^2}{3n\tau_i^2}. \tag{31}$$

A voxel grid does not distinguish the events of a pixel, so the doubled timing error of (28) lowers the fidelity only by a term of order $1/n$.

Equation (31) is the expected fidelity of the events of a given source pixel moved along the true motion. When the bin is shorter than the event spacing, the count of a pixel is zero or one, which bins hold the ones is decided by the phases, and about a share $1-\Delta_t/\tau_i$ of the counts remains a residual. When the bin is long against the spacing, a bin holds whole trains or long parts of them, the offset δ moves only the events at their ends, and the moved counts approach the true ones. Prediction along the true motion therefore removes most of the counts on long bins and few on short ones.

VI. Experiments

The experiments address three questions: on the model, the three regimes of Theorem 2 (Section VI-B); on the model, whether moved counts follow (31) and what temporal prediction costs (Section VI-C); on recordings, whether the phases of the pixels are independent (Section VI-D); and what the mechanism means for a standard video coder on recordings (Section VI-E).

A. Protocol

Data. The recordings from a static camera are the 112 recordings of the training split of eTraM [24], 61 by day and 51 by night, from a sensor of 1280×720 pixels above a road, with 2.4×10^{10} events. The recordings from a moving camera are the 41 training sequences of DSEC [25], left event camera, 640×480 pixels, on a car, with 3.7×10^{10} events in 2634 s. A window of 33.3 ms of the median DSEC sequence holds 1.15 events per pixel, against 0.04 for eTraM. Twelve eTraM recordings and five DSEC sequences are held out for tuning, and the results below are over the other 100 and 36.

Grids. Time is cut into windows of $T_0 = 33.3$ ms and each window into k slices, so $\Delta_t = T_0/k$. A voxel holds $s \times s$ pixels and one polarity. We use $s \in \{1, 2\}$ and $k \in \{1, 3, 6, 12, 32, 64\}$, from $\Delta_t = 33.3$ ms to 0.52 ms. A group is four consecutive windows, taken every 3 s of an eTraM recording and every 2 s of a DSEC sequence.

Video coder. The counts of a group form one clip of $4k$ pictures, with the two polarities side by side in the luma plane of a 4:2:0 picture and constant chroma. They are coded by x265 3.5 [27], an encoder of HEVC [22], through ffmpeg 4.4.2, in the Main profile at the preset `slow`. Each clip has one intra picture and P pictures after it, without B pictures. Adaptive quantization and the psychovisual options are off. In lossless mode the luma is the count. At a constant quantization parameter (QP) the luma is 16 times the count, the decoder rounds the luma divided by 16, and the intra picture is coded at the QP of the P pictures, except for the operating points on the grid of two pixels and $T_0/3$, where it has the default QP offset of x265. Counts are clipped at 15. On one-pixel voxels with $k \ge 3$, clipping removes less than 1% of the events; at $k = 1$ and on two-pixel voxels it removes up to 6% and 28%, by night. The fidelity of a decode is measured against the true events, so clipped events count as lost. Every bit of the HEVC stream is counted, parameter sets included.

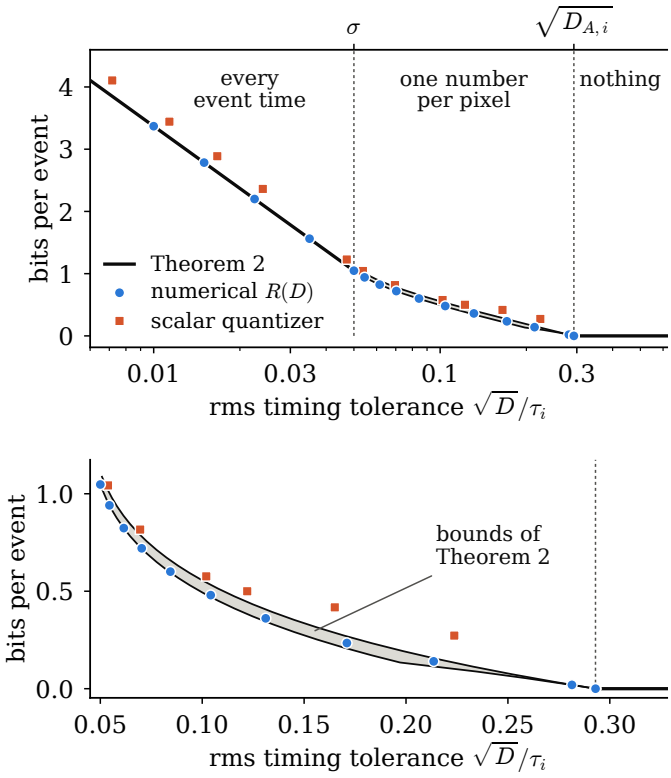


Fig. 2. Rate of one pixel against the timing tolerance for $n = 3$ and $\sigma/\tau_i = 0.05$: Theorem 2, a numerical rate-distortion function and a scalar quantizer. Bottom: the regime between σ and $\sqrt{D_{A,i}}$.

Voxel list. As a coder without prediction, the voxel list sorts the events of a window by voxel, writes the gaps between their voxel indices in a variable-length integer code and compresses the result with Zstandard at level 19 [28]. It codes the same clipped counts as HEVC and returns them exactly.

Tasks. With the annotated objects of eTraM and of DSEC-Detection [29], the fidelity (29) inside the boxes of the objects (box F), the share of boxes that the decode leaves without events, and the error of the motion of each box estimated by contrast maximization [30], in pixels per window.

Statistics. Bits are per true event. A recording enters with the sums over its groups, and we report the median over recordings, or the value pooled over recordings with a 95% interval from a bootstrap over recordings where one is given. The coder comparisons of Section VI-E use one group in 40 of eTraM and one in 20 of DSEC, with all grids and QPs (42 recordings by day, 34 by night and 36 sequences); the representation of Section VI-E and the phase test of Section VI-D use one group in five (55, 45 and 36 recordings).

B. The Three Regimes

Fig. 2 compares Theorem 2 with a rate-distortion function that is computed without it, for one pixel with $n = 3$ and $\sigma/\tau_i = 0.05$. These values illustrate the theorem and are not those of a particular sensor. By (33), the event times are an orthogonal image of the scalar X_i, which carries the phase, and of $n-1$ normal variables. The iteration of [31] gives the rate-distortion function of X_i on a grid of 1217 points, and the normal components enter in closed form at the same slope.

Below σ^2 the numerical rate equals (17) to within 4×10^{-6} nat. Between σ^2 and $D_{A,i}$ it stays inside the band between the bounds (18) and (21), which is at most $\Gamma_i = 0.187$ bit per pixel wide here, against the limit of 0.255 bit. The rate leaves σ^2 on the lower bound $U_i - \Gamma_i$: the phase is still described finely there, and the code profits fully from its distribution, which is uniform and not normal. Toward $D_{A,i}$ the description becomes coarse, the rate depends on the phase only through its variance, and the numerical rate, the bound (21) and U_i meet. From $D_{A,i}$ on the rate is zero, and the regenerated events (22) of 2×10^6 simulated pixels have the distortion $0.9994\,D_{A,i}$.

The squares are a uniform scalar quantizer with entropy coding, applied to what each regime describes. On every component of the event times it stays 0.254 to 0.258 bit per event above (17). This is the loss $\frac{1}{2}\ln(2\pi e/12)$ of a uniform quantizer at fine resolution, the difference between the entropy $h - \ln\Delta$ of the index of a quantizer of step Δ and (17) at the distortion $\Delta^2/12$. On the mean event time alone, with the other components left out, it stays 0.09 to 0.19 bit per event above the lower bound up to $D_{A,i}$. A code that describes every event time, then one number per pixel, then nothing, is therefore within 0.26 bit per event of the limit in all three regimes.

C. Moved Counts on the Model

Straight edges cross a sensor of 320×240 pixels at six pixels per window, 40 pixels apart, for twelve windows. Every row of pixels is a path of the model, with τ from 0.25 to 8 ms, $n \in \{1, 3\}$ and $\sigma = 0.05\,\tau$. The prediction of window m is the counts of window $m-1$ moved by exactly six pixels, the true-motion prediction.

Wherever (31) applies, in all twelve cells of τ, n and k with $\Delta_t \le \tau/2$, the fidelity of the moved counts lies 0.001 to 0.005 below it (Fig. 3a). As Δ_t/τ falls from 33 to 0.52 at $\tau = 1$ ms, the fidelity falls from 0.990 to 0.489, so half of the counts are a residual even under the true motion.

x265 codes the same counts without loss, once with P pictures and once with intra pictures only (Fig. 3b). Its P pictures cost 0.14 bit per event at $\Delta_t/\tau = 33$ and 7.9 at 0.52 for $\tau = 1$ ms. They undercut the voxel list for $\Delta_t/\tau \ge 5.6$, match it near 2.8, and cost up to 1.7 times as much below 1.4. Against intra pictures only, they save a factor of 2 to 2.5 at $\Delta_t/\tau \ge 11$ and of 1.1 to 1.3 below 1. The cost of temporal prediction thus rises where (31) says that moved counts stop predicting the true ones.

D. The Phase on Recordings

On recordings neither the motion nor the event spacing is known, and (31) cannot be evaluated pixel by pixel: many

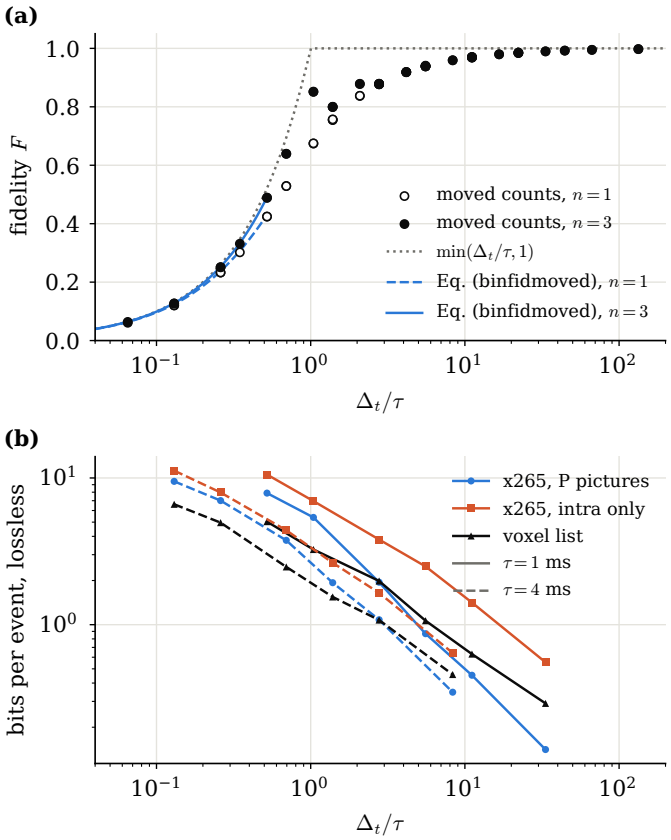


Fig. 3. The model, one-pixel voxels. (a) Fidelity of the counts moved along the true motion. (b) Lossless bits per event for $n = 3$.

pixels fire one or two events per passing edge, so an interval between their events often separates two passages. We test its premise instead, the independence of the phases. For each block of 20×20 pixels the shift that pairs the most events of window $w-1$ with window w is chosen, and the same shift, with the median offset between the first moved and the first true event of each pixel as a time offset, predicts window $w+1$ from window w. For a target pixel and polarity whose true events in window $w+1$ hold two consecutive intervals within a factor 1.2 of each other, τ_i is the median of such intervals, and the position of the moved events in its train is

$$u_i = \frac{1}{2\pi}\arg\Big(\sum_k e^{2\pi\mathrm{i}\,p_k/\tau_i}\sum_l e^{-2\pi\mathrm{i}\,t_l/\tau_i}\Big), \qquad (32)$$

over its predicted times p_k and its true times t_l. Under (A3), $A_j - A_i$ is uniform modulo τ_i, so u_i is uniform on $[0, 1)$ whatever the residual offset of the motion and the length of the trains, and the mean of $\cos 2\pi u_i$ is 0. If the pixels of a path shared their phases, u_i would be constant. On the model at 1.5 to 13.3 pixels per window, shared phases give a mean of 0.91 to 0.97, and independent phases a mean within four standard errors of 0. Before the recordings were read, a 95% interval below 0.1 was set to read as independent phases, one above 0.3 as correlated phases, and anything else as partial correlation. The test uses one group in five, 1264 groups, and the moving blocks; blocks with a zero shift are left out, since a pixel moved onto itself keeps its own phase.

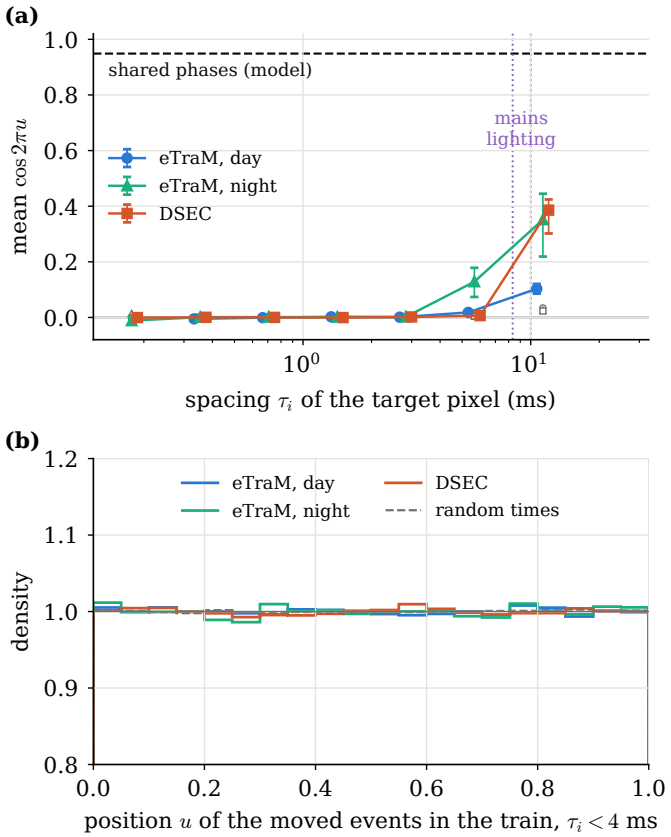


Fig. 4. Recordings, moving blocks. (a) Mean of $\cos 2\pi u$ against the spacing of the target pixel, with 95% intervals; grey: random times. (b) Histogram of u for $\tau_i < 4$ ms.

The phase of moved events is uniform wherever the trains come from motion (Fig. 4). For pixels with τ_i between 0.25 and 4 ms, 84% of the tested pixels by day, 62% at night and 62% on DSEC, the mean of $\cos 2\pi u_i$ lies within 0.006 of 0 in every bin of τ_i, against 0.91 to 0.97 for shared phases, and the histogram of u_i is flat. Pooled over all spacings, the mean is 0.007 (95% interval 0.005 to 0.008) by day, which reads as independent, and 0.077 (0.044 to 0.106) at night and 0.096 (0.042 to 0.148) on DSEC, which read as partial. The partial readings come from pixels with τ_i between 8 and 16 ms, where the mean reaches 0.35 at night and 0.39 on DSEC. That band holds the periods of mains lighting, 8.33 and 10 ms, which modulate many pixels in step, a common cause outside the model.

The fidelity of the same prediction, binned by Δ_t/τ_i with τ_i the median regular interval of the pixel over 133 ms, is 0.142 by day and 0.151 on DSEC at $\Delta_t/\tau_i \approx 1/8$ on the grid of one pixel and $T_0/32$, against 0.141 from (31), and 0.190 at night. Only 30% to 48% of the events lie in such pixels, and the remaining events come from pixels without a regular train.

E. A Standard Video Coder

HEVC on count pictures has been used to code event streams without loss [11]. We use x265 as a probe of what the mechanism means for such a coder: it has motion-compensated prediction, a transform and a quantizer, and it records how it codes each picture.

TABLE I
LOSSLESS BITS PER EVENT: X265 WITH P PICTURES (P), WITH INTRA PICTURES ONLY (I), AND THE VOXEL LIST (Q).

	eTraM, day			eTraM, night			DSEC		
(s,k)	P	I	Q	P	I	Q	P	I	Q
$(1,1)$	2.13	2.46	2.18	1.97	2.19	2.01	1.64	1.70	1.69
$(1,3)$	3.32	4.81	3.40	3.14	3.90	3.25	2.76	3.00	2.83
$(1,12)$	6.43	11.3	6.12	5.51	8.17	5.59	4.88	5.12	4.74
$(1,64)$	11.5	38.2	10.2	9.75	24.6	9.06	8.09	8.39	7.48
$(2,3)$	2.08	2.70	2.17	1.93	2.29	1.99	1.49	1.69	1.59

Without loss. x265 with P pictures needs about the bits of the voxel list, which predicts nothing, and more on fine grids (Table I). Over the twelve grids, the ratio of the two is 0.92 to 1.02 at slices of $T_0/6$ or longer, 1.00 to 1.07 at $T_0/12$, and 1.06 to 1.18 at $T_0/32$ and $T_0/64$. On DSEC, where the camera moves, the same x265 with intra pictures only needs 1.03 to 1.10 times the bits of the P pictures at every slice, so temporal prediction saves little there. On eTraM the intra pictures cost 1.1 to 3.4 times more, increasingly so at fine slices, where they also cost up to 3.7 times the voxel list: a static camera leaves pictures almost empty, and the P pictures save the cost of signaling empty regions, which an intra picture pays anew. Neither version of x265 undercuts the voxel list on fine grids, as Theorem 1 and (31) lead one to expect: on fine grids the counts are set by the phases, which the motion does not carry. Coding the luma as 16 times the count, the mapping of the lossy modes, costs 1.9 to 2.5 times the lossless bits by itself.

With loss. On a fine grid almost all nonzero counts are one: 92% to 99% of the input events lie in voxels of one event at $T_0/64$. The quantizer then removes events instead of coarsening them (Fig. 5b). At $T_0/64$ and QP 20 or below, 91% to 99% of the errors of a decode are voxels of one event that it empties, and 1% to 7% are events that it adds. At QP 20 by day the decode removes 27% of the input events, and at QP 28 it removes 90%. The P pictures lose more of them than intra pictures at the same QP, so at equal fidelity the intra pictures are as cheap or cheaper: to reach $F = 0.9$ at $T_0/64$ they need 1.00, 0.89 and 0.72 times the bits of the P pictures by day, at night and on DSEC, and 0.88, 0.78 and 0.69 times to reach 0.95. The box fidelity follows (Fig. 5c): at one bit per event by day the best point at $T_0/64$ keeps a box fidelity of 0.011 and leaves 71% of the boxes without events, against 0.279 and 38% at $T_0/12$. On coarse grids the quantizer coarsens counts that hold several events. On the grid of two pixels and $T_0/3$, x265 by day reaches $F = 0.975$ with 3.26 bits per event at QP 14, 0.933 with 1.09 at QP 28, 0.840 with 0.31 at QP 36 and 0.706 with 0.051 at QP 46. These counts are clipped at 15, which costs up to 0.16 of fidelity at night, where up to 28% of the events lie above the clip, and at most 0.04 of box fidelity.

Representation. The grid itself loses information before any coding. Turning the events into counts and back, with the events of a voxel placed at its first instant, loses nothing for the box fidelity at one pixel and at least three slices per window,

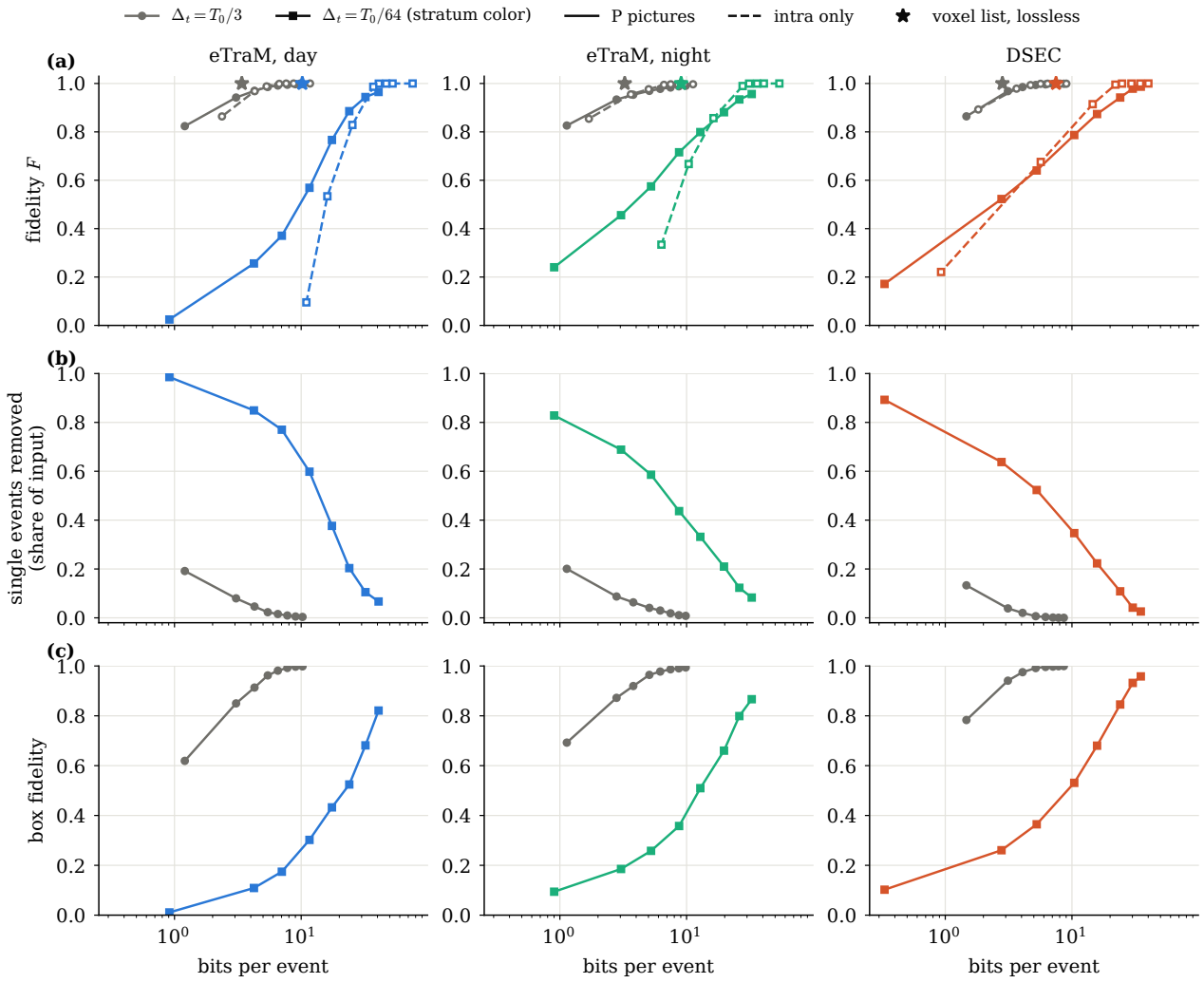


Fig. 5. x265 on count pictures of recordings, one-pixel voxels, $\Delta_t = T_0/3$ (grey) and $T_0/64$. (a) Fidelity, (b) single events removed by the decode, (c) box fidelity, against bits per event.

with no box left empty. Two-pixel voxels raise the motion error: at $k = 3$ it is 4.01, 1.17 and 2.67 pixels per window by day, by night and on DSEC, against 1.49, 0.67 and 0.94 at one pixel.

Design rule. The bin of the receiver decides what motion compensation can do, through the ratio Δ_t/τ. When the bin is long against the event spacing, moved counts predict the true ones, voxels hold several events, and the prediction and the quantizer of a video coder both work. When it is short, the phases of the pixels decide the counts, and on recordings these phases are independent from pixel to pixel. A sender for such a receiver gains nothing from motion-compensated prediction of counts, a list of the counts is the cheaper lossless code, and a lossy code should describe event times relative to the prediction rather than quantize amplitudes.

VII. Conclusion

The pixels on the path of a moving edge share the edge and its motion, and each has a threshold phase and a jitter of its own, which a lossless code must describe at every pixel. A timing tolerance removes first the jitter and then the phase from what a code must describe, and beyond a threshold set by the event spacing the events are regenerated from the edge and the motion alone. A sender that has only the events can move them along the motion, which doubles the timing error and, on a voxel grid, reproduces a share Δ_t/τ of the counts when the bin is shorter than the event spacing. On recordings from a static and a moving camera, the moved events fall at uniform positions within the trains of their target pixels, as independent phases imply, except in the band of mains lighting. A standard HEVC coder on count pictures shows the consequences: on fine grids its P pictures do not undercut a list of the counts, and its lossy errors are single events that it removes, so that intra pictures reach a given fidelity with as few bits or fewer. The bin of the receiver against the event spacing therefore tells a sender whether motion-compensated prediction of counts can pay. The theory covers one ramp edge in uniform passage, so texture, noise events and threshold mismatch remain to be modeled.

APPENDIX A
PROOFS OF LEMMAS 1 AND 2

Proof of Lemma 1. The reference starts in $a_i + C\mathbb{Z}$ by the definition of the phase. By (A1) it changes by $\pm C$, so it stays in this set, and an event requires $\ell_i = \mu_i \pm C$. □

Proof of Lemma 2. By (A2) and (A2′), ℓ_i is nondecreasing, equal to 0 before t_i, to $\gamma r_i(t - t_i)$ during the passage, and to V after it. Since $\mu_i \le 0 = \ell_i$ before the edge arrives and $\mu_i = a_i - C$, the events occur when ℓ_i reaches the levels $a_i + (k-1)C$, $k = 1, 2, \dots$, as long as the level is at most $V = nC$. Because $0 < a_i \le C$, exactly the levels $k = 1, \dots, n$ are reached. Solving $\gamma r_i(t - t_i) = a_i + (k-1)C$ gives (10), and a_i uniform on $(0, C]$ makes A_i uniform on $(0, \tau_i]$. □

APPENDIX B
PROOF OF THEOREM 1

By Lemma 2, $T_i = c_i + A_i\mathbf{1} + \varepsilon_i$ with c_i a function of the template and the flow, $\mathbf{1}$ the all-ones vector, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_n)$. Independence over i follows from (A3) and (A4). Let Q be an orthogonal matrix whose first row is $\mathbf{1}^\top/\sqrt{n}$. Then $Q(T_i - c_i) = (X_i, W_i)$ with

$$X_i = \sqrt{n}\, A_i + \varepsilon_i^{\parallel}, \qquad W_i \sim \mathcal{N}(0, \sigma^2 I_{n-1}), \tag{33}$$

where $\varepsilon_i^{\parallel} \sim \mathcal{N}(0, \sigma^2)$. The variables X_i and W_i are independent, because $Q\varepsilon_i$ has independent coordinates and A_i is independent of ε_i. Differential entropy is invariant under translations and orthogonal maps, so $h(T_i) = h(X_i) + h(W_i)$, which is (11).

Let f be the density of T_i and f_δ its average over the cells of the clock lattice. A cell has probability $\delta^n f_\delta$, so $H(Y_i \mid g, \phi) = n\ln(1/\delta) - \int f \ln f_\delta$ and $e_\delta = \int f \ln(f/f_\delta)$ is a relative entropy, which is nonnegative. The density f is a mixture over A_i of $\mathcal{N}(m, \sigma^2 I_n)$ densities and f_δ is the same mixture of their cell averages. By the joint convexity of relative entropy, e_δ is at most its largest value for a single $\mathcal{N}(m, \sigma^2 I_n)$ density, which is n times the value for a scalar normal density and depends only on δ/σ and on the offset of m in a cell. It tends to zero with δ/σ, uniformly in the offset [26, Ch. 8].■

APPENDIX C
PROOF OF THEOREM 2

The distortion (14) is invariant under the translation by c_i and the orthogonal map Q, so R_i is the rate-distortion function of the independent components (X_i, W_i) of (33) under the total squared error nD. Their variances are s_i^2 and σ^2, the latter $n-1$ times. We drop the index i and write η for $\eta(\sqrt{n}\,\tau_i, \sigma)$ and $\ln^+ x = [\ln x]^+$.

Lower bound (17) *and* (18). Let $(\hat{X}, \hat{W})$ have component errors d_X and $d_1, \dots, d_{n-1}$ with $d_X + \sum_j d_j \le nD$. Independence of the components gives $I(X, W; \hat{X}, \hat{W}) \ge I(X; \hat{X}) + \sum_j I(W_j; \hat{W}_j)$. For a Gaussian component, $I(W_j; \hat{W}_j) \ge \frac{1}{2}\ln^+(\sigma^2/d_j)$, and $I(X; \hat{X}) \ge h(X) - h(X - \hat{X}) \ge \eta - \frac{1}{2}\ln(2\pi e\, d_X)$. The function $\ln^+(\sigma^2/d)$ is convex in d, so the Gaussian terms are smallest when all d_j equal a common d_W, and $d_W > \sigma^2$ brings no benefit. For $n \ge 2$ this gives

$$\begin{aligned} R(D) \;\ge\; \min_{d_W} \Big[\eta - \tfrac{1}{2}\ln\big(2\pi e\,(nD - (n-1)d_W)\big) \\ + \tfrac{n-1}{2}\ln\frac{\sigma^2}{d_W} \Big], \end{aligned} \tag{34}$$

with the minimum over $0 < d_W \le \sigma^2$, $d_W < nD/(n-1)$. The bracket is convex in d_W and its derivative vanishes only at $d_W = D$. The minimum is at $d_W = D$ when $D \le \sigma^2$, where it equals the right side of (17), and at $d_W = \sigma^2$ otherwise, where it equals $U(D) - \Gamma$. For $n = 1$ there is no W and the same values follow with $d_X = D$.

Equality for $D \le \sigma^2$. Write $\varepsilon = \varepsilon' + Z$ with independent $\varepsilon' \sim \mathcal{N}(0, (\sigma^2 - D)I_n)$ and $Z \sim \mathcal{N}(0, DI_n)$, and let $\hat{T} = c + A\mathbf{1} + \varepsilon'$. Then $T = \hat{T} + Z$ with Z independent of $\hat{T}$, the distortion is D, and $I(T; \hat{T}) = h(T) - h(Z)$, which is the lower bound.

Upper bound for $\sigma^2 < D < D_A$. Spend no rate on W, at error σ^2 per component, and describe X at error $d = nD - (n-1)\sigma^2 \in (\sigma^2, s^2)$ through $\hat{X} = (1 - d/s^2)(X - \mathbb{E}X + Z') + \mathbb{E}X$, with Z' an independent normal variable of variance $s^2 d/(s^2 - d)$. The error is d, and $I(X; \hat{X}) = I(X; X + Z') \le \frac{1}{2}\ln(s^2/d)$, because a normal input maximizes the mutual information across an additive normal noise channel at a given input variance [26].

Lower bound (21) *and zero rate.* For a variable Ξ with $\ln \mathbb{E}\, e^{\lambda(\Xi - \mathbb{E}\Xi)} \le \lambda^2\varsigma^2/2$ for all λ under a law P, and any law Q, the nonnegativity of the relative entropy between Q and the law with density proportional to $e^{\lambda\Xi}$ with respect to P gives $D(Q\|P) \ge \lambda(\mathbb{E}_Q\Xi - \mathbb{E}_P\Xi) - \lambda^2\varsigma^2/2$, and the best λ gives $(\mathbb{E}_Q\Xi - \mathbb{E}_P\Xi)^2 \le 2\varsigma^2 D(Q\|P)$. With P the law of X and Q its conditional law given $\hat{T}$, averaging over $\hat{T}$ yields $\operatorname{Var}\mathbb{E}[X \mid \hat{T}] \le 2\varsigma^2 I(X; \hat{T})$. The uniform part of X has width $a = \sqrt{n}\,\tau$ and satisfies the condition with $a^2/12$, since $\ln(\sinh x/x) \le x^2/6$, so X satisfies it with $\varsigma^2 = s^2$, and each W_j with $\sigma^2 \le s^2$. Since the conditional mean minimizes the squared error and the components are independent,

$$\begin{aligned} nD &\ge \mathbb{E}\|T - \mathbb{E}[T \mid \hat{T}]\|^2 \\ &= nD_A - \operatorname{Var}\mathbb{E}[X \mid \hat{T}] - \sum_j \operatorname{Var}\mathbb{E}[W_j \mid \hat{T}] \\ &\ge nD_A - 2s^2 I(T; \hat{T}), \end{aligned} \tag{35}$$

which is (21). With no description the best output is the mean, which is (22), at distortion $\frac{1}{n}(s^2 + (n-1)\sigma^2) = D_A$, so (19) holds. ■

APPENDIX D
PROOF OF THEOREM 3

The vectors T_i are independent and the distortion is additive over i. For any $\hat{T}$ with average distortion at most D, let $D_i = \mathbb{E}\, d(T_i, \hat{T}_i)$. Then $I\big((T_i)_i; (\hat{T}_i)_i\big) \ge \sum_i I(T_i; \hat{T}_i) \ge \sum_i R_i(D_i)$ with $\frac{1}{M}\sum_i D_i \le D$. Conversely, independent test channels whose rates are within any given margin of $R_i(D_i)$ come within M times that margin of the sum. Each R_i is convex, nonincreasing and continuous, and tends to infinity as $D_i \to 0$ by (17), so the minimum in (23) exists.

For $D < \sigma^2$, every R_i is differentiable at D with the derivative $-n/(2D)$ by (17), and it is convex on $(0,\infty)$. Hence $R_i(D_i) \geq R_i(D) - \frac{n}{2D}(D_i - D)$ for every $D_i > 0$, and summing over i with $\sum_i (D_i - D) \leq 0$ shows that the allocation $D_i = D$ is optimal. This gives (24) for $D < \sigma^2$, and the case $D = \sigma^2$ follows by continuity.

By Theorem 2, $\sum_i R_i(D_i) = 0$ if and only if $D_i \geq D_{A,i}$ for every i, and such an allocation with average at most D exists if and only if $D \geq \frac{1}{M}\sum_i D_{A,i} = \bar{D}_A$. Finally, (21) and $s_i^2 \leq n\tau_{\max}^2/12 + \sigma^2$ give $\sum_i R_i(D_i) \geq n\sum_i [D_{A,i} - D_i]^+/(2(n\tau_{\max}^2/12 + \sigma^2))$, and $\sum_i [D_{A,i} - D_i]^+ \geq M(\bar{D}_A - D)$, which is (26). ∎

Appendix E
Proof of Proposition 1

Divide side j of the box into the smallest number of equal cells of length at most $\varsigma_j = \sqrt{\xi}\,\sigma/(q\,S_j M^{\beta_j})$, and let $\hat{\theta}$ be the center of the cell that contains θ. Side j has at most $L_j M^{\alpha_j}/\varsigma_j + 1$ cells, so the index of $\hat{\theta}$ takes $\sum_j \ln\big(L_j S_j q\, M^{\alpha_j+\beta_j}/(\sqrt{\xi}\,\sigma) + 1\big)$ nats, which is (27). The decoder outputs $\hat{T}_{i,k} = \bar{t}_{i,k}(\hat{\theta})$. The segment from θ to $\hat{\theta}$ lies in the box, so by the mean value theorem $e_{i,k} = \bar{t}_{i,k}(\theta) - \bar{t}_{i,k}(\hat{\theta})$ satisfies $|e_{i,k}| \leq \sum_j S_j M^{\beta_j}\varsigma_j/2 \leq \sqrt{\xi}\,\sigma/2$. By (10), $T_{i,k} - \bar{t}_{i,k}(\theta) = A_i - \tau_i/2 + \varepsilon_{i,k}$ has mean zero given θ and variance $D_{A,i}$, and $e_{i,k}$ is a function of θ. Hence $\mathbb{E}\,(T_{i,k} - \hat{T}_{i,k})^2 = D_{A,i} + \mathbb{E}\,e_{i,k}^2 \leq D_{A,i} + \xi\sigma^2/4$, and the average over the path is at most $\bar{D}_A + \xi\sigma^2/4 \leq (1+\xi)\bar{D}_A$. ∎

Appendix F
Proof of Proposition 2

By (10) and (A4), $T_{i,k} - \tilde{T}_{i,k} = (A_i - A_j) + (k-1)(\tau_i - \tau_j) + \varepsilon_{i,k} - \varepsilon_{j,k}$. The phases and the jitter of the two pixels are independent, so this difference has the mean $(k - \frac{1}{2})(\tau_i - \tau_j)$ and the variance $\tau_i^2/12 + \tau_j^2/12 + 2\sigma^2 = D_{A,i} + D_{A,j}$. Its mean square is the variance plus the squared mean, and $\frac{1}{n}\sum_{k=1}^{n}(k - \frac{1}{2})^2 = (4n^2 - 1)/12$. ∎

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